GlycoTwin-AI: Multimodal Digital Twin for Proactive Diabetes Management

Happiest Health Reimagining & Reforming Healthcare in India Summit 2026, Bengaluru

- Track: Digital Twin Proof-of-Concept (PoC) Challenge
- Focus Condition: Type 2 Diabetes Mellitus & Postprandial/Nocturnal Dysglycemia
- Target Population: Asian Indian Phenotype (>101M Diabetics, high carb staple diets)
- Core Innovation: In-silico physiological twin fusing static EHR with wearable CGM/PPG
- Key Capability: 2-Hour ahead adverse event prediction & What-If counterfactual simulation

The Indian Diabetes Crisis: Why Reactive Healthcare Fails

ICMR-INDIAB 2023 Epidemiological Context & Clinical Motivation

- 101.3 Million Diabetics + 136 Million Pre-diabetics in India ('Diabetes Capital of the World')
- Asian Indian Phenotype: High visceral fat & severe insulin resistance at lower BMI (>= 23 kg/m²)
- Dietary Trap: Refined rice/wheat staples cause massive postprandial glucose surges (>200 mg/dL)
- Silent Danger: Nocturnal hypoglycemia (<70 mg/dL) during sleep causes fatal cardiac arrhythmias
- Reactive Failure: Quarterly HbA1c tests are 3-month trailing averages; they miss intraday volatility
- Solution: Continuous Digital Twin providing real-time in-silico simulation & proactive alerts

Dual Data Stream Fusion Architecture

Synthesizing Static Clinical Records with High-Frequency Wearable IoT Telemetry

- Stream 1 (Static EHR): Demographics, Asian Indian BMI, baseline HbA1c, eGFR, lipid profile,
- comorbidities (retinopathy, neuropathy, CVD), TCF7L2 genetic risk, & baseline medications.
- Stream 2 (Dynamic IoT Wearables): 5-minute CGM interstitial glucose, continuous Heart Rate (bpm),
- HRV (RMSSD autonomic tone), step cadence, sleep architecture stages, & Indian meal carbs (g).
- Feature Synthesis: 42 kinetic & cross-domain features including velocity (dG/dt), acceleration (d^2G/dt^2),
- intraday CV% volatility, and autonomic stress coupling (HR / HRV ratio).
- Compliance: 100% compliant with Indian DPDP Act 2023 & HIPAA sandbox regulations.

Dual-Head Machine Learning Core & Clinical Metrics

Clinical Validation on Unseen Patient Test Cohorts (Zero Patient Leakage)

Continuous Regressor (2-Hour Horizon Forecast):

- MARD (Mean Absolute Relative Difference): 7.15% (Gold standard clinical threshold < 10%)
- RMSE: 14.50 mg/dL | MAE: 11.11 mg/dL across 59,940 high-resolution multimodal records
- Generates dynamic 95% confidence bounds reflecting biological uncertainty

Adverse Event Classifier (Early Warning Lead Time: 120 minutes):

- Severe Postprandial Hyperglycemia Spike (>200 mg/dL): Precision = 65.31%
- Impending Nocturnal Hypoglycemia (<70 mg/dL): Calibrated safety decision threshold (18%)
- Macro F1: 0.54 | Patient-stratified holdout evaluation

Physician 'What-If' In-Silico Counterfactual Simulator

Empowering Doctors to Test Interventions Digitally Before Clinical Execution

- Clinical Need: Physicians cannot conduct risky trial-and-error experiments on living patients.
- Digital Twin Sandbox: Simulates three high-impact clinical interventions in real time (<50ms):
    1. Post-Prandial Brisk Walk (0 to 45 mins): Simulates muscular GLUT-4 glucose clearance.
    2. Indian Carb Swap / Reduction (0 to 60g): Models switching white rice to millets/proteins.
    3. Medication Adjustment (-4U to +4U): Titrates basal insulin or shifts sulfonylurea timing.
- Instant Feedback: Re-renders simulated counterfactual trajectory vs. baseline projection curve,
- calculating exact projected glucose delta (mg/dL) and RSSDI/ADA clinical recommendation.

Clinician Digital Twin Cockpit UI

Human-in-the-Loop Web Platform for Proactive Patient Management

- Full-Featured Glassmorphic Web App: Built with FastAPI, Tailwind CSS, & Chart.js.
- Live Telemetry Waveforms: Synchronized visualization of CGM, Heart Rate, HRV, steps, & meals.
- Triage Early Warning System: Color-coded alert badges (Green = Stable, Amber = Hyper, Red = Hypo).
- Explainable AI (XAI) Drawer: Physiological root-cause attributions explain why alerts fired.
- Interactive Simulator Panel: Direct slider controls for walking, carb reduction, and insulin titration.
- Zero-Build Deployment: Runs instantly out-of-the-box on localhost or cloud servers.

Clinical Walkthrough: Two Real-World Scenarios

Proactive Intervention in Indian Patient Personas

Scenario A (Postprandial Hyperglycemia - Patient IND-PT-10007):

- 54y male, Rice staple diet, baseline HbA1c 9.4%. Post-lunch predicted spike: 218 mg/dL.
- Doctor simulates: 20-min post-lunch walk + 35g carb reduction (foxtail millet swap).
- Twin outcome: Predicted 2h glucose drops by -38 mg/dL, restoring safe in-range control (180 mg/dL).

Scenario B (Impending Nocturnal Hypoglycemia - Patient IND-PT-10004):

- 59y female on basal insulin (22U). Twin detects 02:30 AM autonomic crash (HRV drops to 14ms).
- Twin flags impending 04:30 AM hypo crash (61 mg/dL) 2 hours in advance.
- Doctor simulates: Titrate evening basal insulin by -3U + bedtime complex protein snack -> Crash averted.

DPDP Act 2023 & HIPAA Compliance Architecture

Patient Privacy, Synthetic Data Integrity & Regulatory Sandbox Rules

- Indian DPDP Act 2023 Compliance: Fully synthetic patient cohorts adhering to national privacy standards.
- Zero Real Patient PII: Synthetic identities (IND-PT-XXXXX) generated with authentic clinical distributions.
- Synthea & MIMIC-IV Standard Schema: Statistically verified against ICMR-INDIAB epidemiological metrics.
- Data Leakage Protection: Strict patient-level train/test separation prevents data memorization.
- Audit Logging: Every in-silico simulation and decision recommendation is timestamped and cryptographically logged.

Socioeconomic Impact & Indian Healthcare Deployment Roadmap

From University Incubator Prototype to Ayushman Bharat Digital Mission (ABDM)

- Economic Impact: Avoids ₹40,000 - ₹1,50,000 emergency admissions for DKA and severe hypoglycemia.
- Long-Term Prevention: Blunting glycemic spikes reduces HbA1c by 0.8-1.2%, cutting microvascular complications by 37%.
- Integration Roadmap:
  - Phase 1 (Now): Working Digital Twin PoC, multimodal predictive engine & doctor cockpit.
  - Phase 2 (Summit Final): Hardware integration with commercial CGMs (Abbott FreeStyle Libre) & wearables.
  - Phase 3 (Clinical Pilot): FHIR API integration into Ayushman Bharat Digital Mission (ABDM) & e-Sanjeevani.
  - Phase 4 (Edge AI): Mobile companion app providing real-time personalized patient nudges.

Conclusion & Competitive Summary

Happiest Health Reimagining and Reforming Healthcare Summit 2026

- Accomplished: Selected #1 Indian healthcare priority (Type 2 Diabetes) & delivered an end-to-end Digital Twin.
- Dual Data Fusion: Seamlessly unites static clinical EHR with high-resolution wearable telemetry.
- Clinical Grade Accuracy: 7.15% MARD and 120-minute lead time on acute dysglycemic adverse events.
- Doctor-Centric Innovation: In-silico 'What-If' counterfactual engine transforms reactive care into proactive prevention.
- Deliverables: Complete open-source codebase (MIT), interactive UI, architecture diagrams, and 20-min demo video.
- Vision: Building the proactive foundation for Indian digital healthcare.